SECTION 2 · SUBSECTION 2.1
SYSTEM ARRANGEMENT

2.1.1 INTRODUCTION.

Groundwork turns a field-service member's text and photos into a priced quote, scheduled work, and an invoice. The model interprets intent; code identifies the caller and computes every amount (figure 2.1-1).

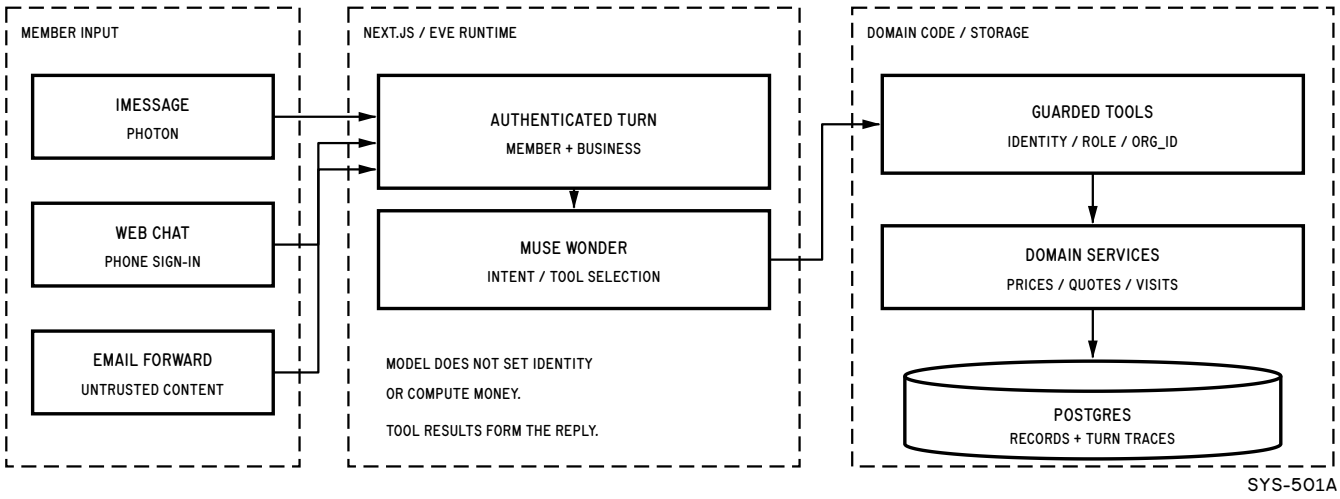


Figure 2.1-1. Member-to-Record Functional Arrangement

2.1.2 FUNCTIONAL DESCRIPTION.

iMessage, web chat, and member-forwarded email reach the assistant. Tools load the business, customer, price book, and notes. An owner or admin approves the quote; the customer signs its current revision on the link. Work is booked separately. An invoice copies the agreed quote (figure 2.1-2).

2.1.2.1 Authority boundary.

Active membership, role, suspension, and business scope are checked in code. Forwarded email may create drafts; it cannot authorize protected changes.

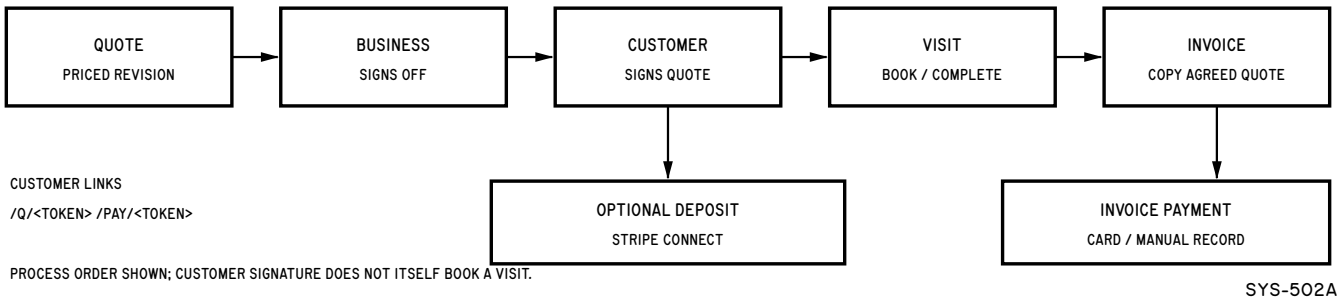


Figure 2.1-2. Quote-to-Payment Operating Sequence

2.1.3 REFERENCE DATA.

Table 2.1-1. System Reference

Parameter	Value
Member roles	owner / admin / member
Data scope	org_id on domain records
Quote history	Immutable revisions
Languages	English / Spanish
Quote validity, base / landscape	30 days / 14 days
Deposit, base / landscape	0% / 30%
Customer acceptance	Current priced revision

CAUTION

An email's contents are not the member's permission to send, approve, or change saved records.

```
NEW JOB                                  MEMBER
1 STATE WORK - text + photos
2 VERIFY CUSTOMER - use records
3 CONFIRM SCOPE - fill gaps
4 REVIEW QUOTE - check PDF
5 APPROVE - owner or admin
6 SHARE - customer link
7 BOOK - crew + day + time
```

SYS

SECTION 2 · SUBSECTION 2.2
QUOTATION AND PHOTO SYSTEM

2.2.1 INTRODUCTION.

A priced photo locates quoted work; it does not measure it. Muse Wonder proposes scope, the member confirms photo-derived quantities, and the calculator prices the quote. SAM 3.1 supplies outlines only (figure 2.2-1).

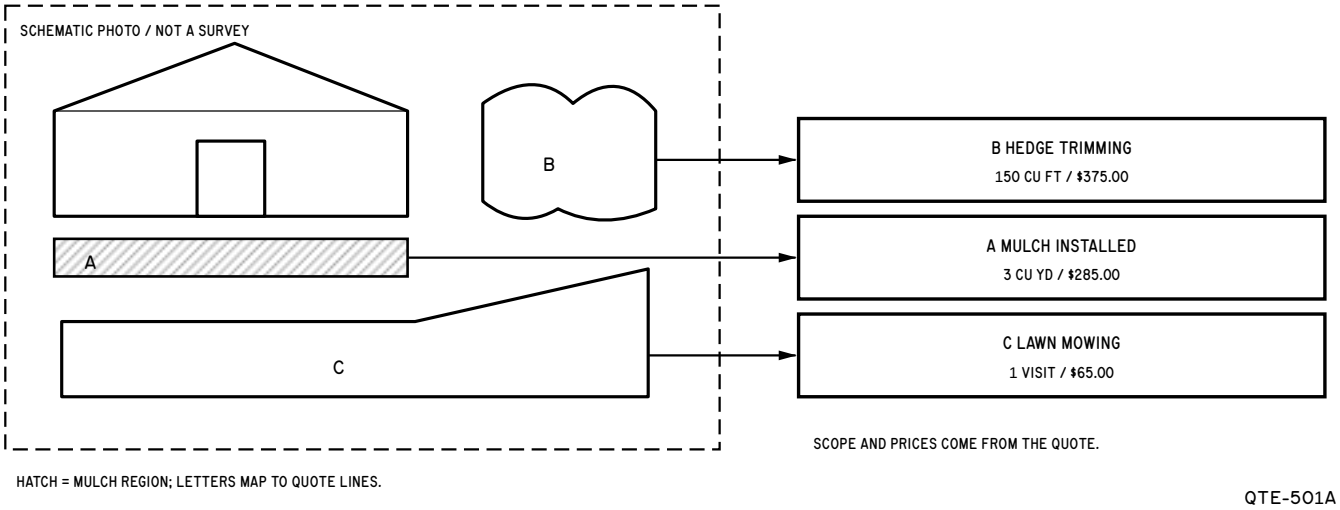


Figure 2.2-1. Photo Regions Assigned to Priced Quote Lines

2.2.2 FUNCTIONAL DESCRIPTION.

The tools resolve saved item IDs to rates. The calculator rounds each line to cents, adds selected rules, applies the minimum, then adjusts and taxes the result (figure 2.2-2). The PDF and photo labels reuse those amounts. A missing price normally returns “needs_owner”; a one-time total override is explicit.

2.2.2.1 Retained segmentation.

SAM masks and candidates are stored. Selecting saved outlines or drawing by hand needs no new SAM call; a box prompt or correction points request segmentation again.

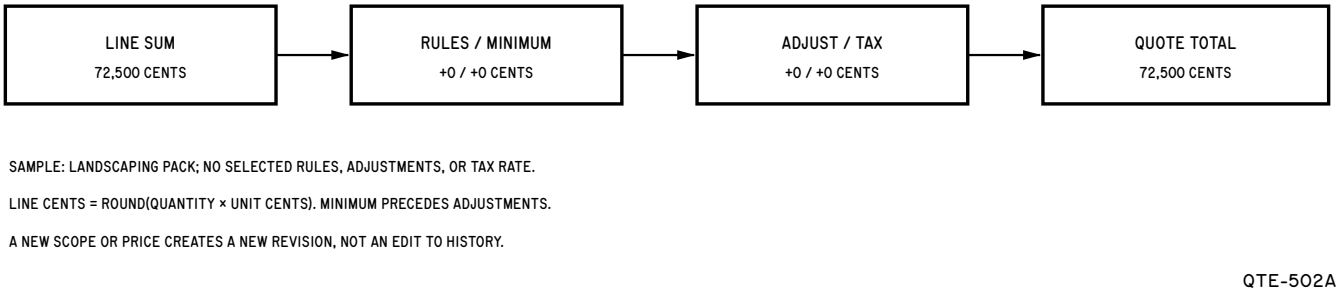


Figure 2.2-2. Calculator Data Path — Worked Landscape Example

2.2.3 REFERENCE DATA.

Table 2.2-1. Verified Pricing and Image Data

Parameter	Value
Money / percentage units	Integer cents / basis points
Mulch installed, per cubic yard	9,500 cents
Hedge trimming, per cubic foot	250 cents
Lawn mowing, per visit	6,500 cents
Landscape minimum charge	15,000 cents
SAM input, maximum long side	1,600 pixels
Photo minimum width	320 pixels

CAUTION

An outline is not a measurement. Count each quoted line once even when it appears in several photos.

QUOTE CHECK	MEMBER
1 CUSTOMER - match or save	
2 ITEMS - use saved names	
3 QUANTITIES - confirm from photo	
4 PRICES - use tool amounts	
5 RULES - select applicable	
6 PHOTO - check missing regions	
7 REVISE - preserve history	

QTE

SECTION 2 · SUBSECTION 2.3
SCHEDULING AND DISPATCH

2.3.1 INTRODUCTION.

Groundwork computes a crew's route from booked visits, the yard, equipment, and work hours. Open slots rank days under capacity first, then added driving. Re-planning proposes changes; it does not silently move work (figure 2.3-1).

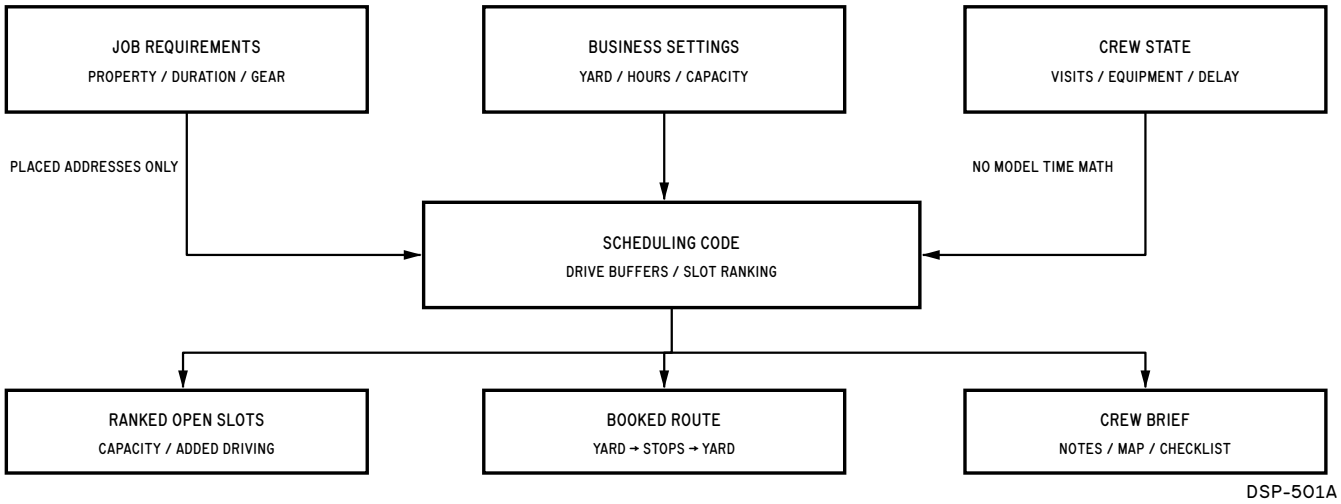


Figure 2.3-1. Availability and Crew-Route Computation

2.3.2 FUNCTIONAL DESCRIPTION.

Crew text "next", "here", "done", or "running 30 late". Tools return the stop, notes, checklist, and revised day plan. Completing a due visit's last checklist item can mark it done. A late day or forecast risk produces a numbered proposal; an owner or admin decides, and code rechecks it (figure 2.3-2).

2.3.2.1 Drive estimate.

Drive time uses straight-line distance × road factor at town speed, rounded up to 5 minutes. No road graph or live traffic is used; unplaced addresses use a stated fallback.

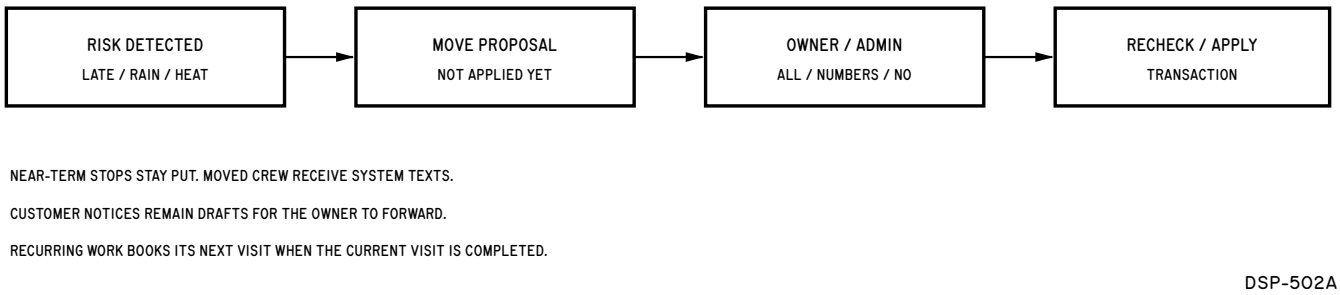


Figure 2.3-2. Guarded Rescheduling Sequence

2.3.3 REFERENCE DATA.

Table 2.3-1. Base Dispatch Settings — Owner-Adjustable

Parameter	Value
Road factor / town speed	1.3 / 28 mph
Day capacity, work + driving	95% of working hours
Running-long threshold	30 min past stop finish
Return-time grace	15 min past workday end
Protected near-term starts	Within 45 min
Rain / heat trigger, inclusive	60% / 95°F
Weather review lead	15 h before workday

NOTE

A shorter route is advice, not an automatic reorder. Proposal customer messages are drafts.

```
RUN THE DAY                    CREW
1 NEXT - read notes + map
2 HERE - start the stop
3 TICK - completed work items
4 PHOTO - attach finished work
5 DONE - close the visit
6 DELAY - report minutes late
7 OWNER - decides any moves
```

SECTION 2 · SUBSECTION 2.4
SETUP AND OPERATING LIMITS

2.4.1 INTRODUCTION.

One Next.js deployment serves multiple businesses with Postgres records and Blob files. Eve hosts the assistant and optional page designer. Fake adapters replace delivery services, not the model or database (figure 2.4-1).

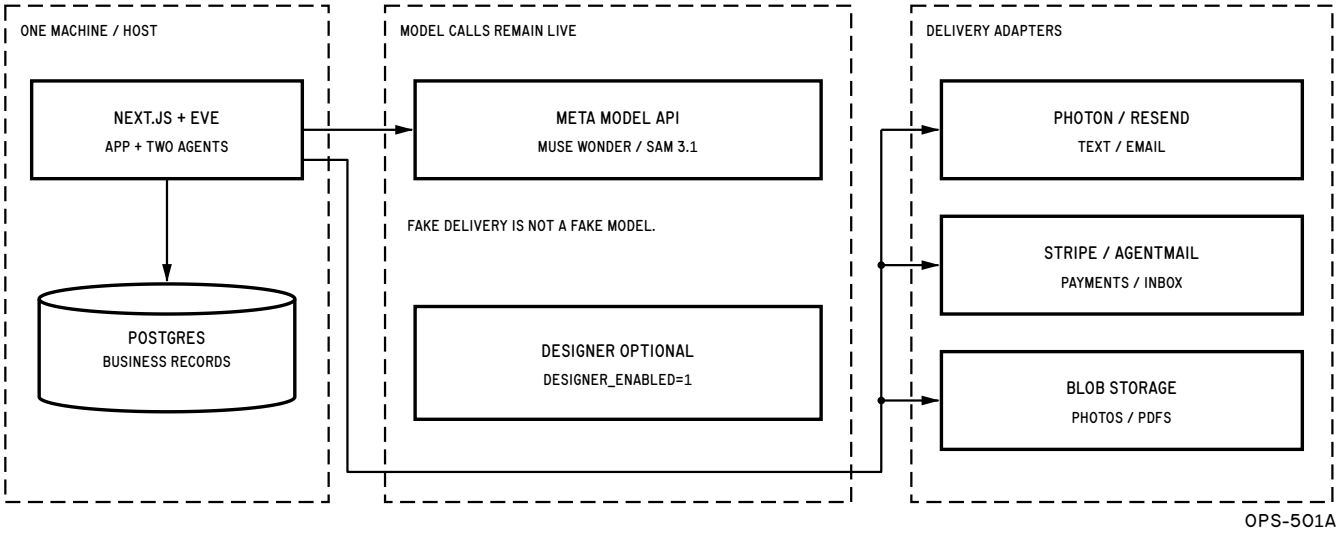


Figure 2.4-1. Runtime and External-Service Arrangement

2.4.2 FUNCTIONAL DESCRIPTION.

Development needs Node, an empty Postgres database, a Meta key, and an auth secret. Migrate, provision the business, seed a trade pack, then start the app. Production also needs public file storage and scheduled jobs. The optional designer changes customer-page layout while code supplies prices and forms (figure 2.4-2).

2.4.2.1 Safe operating modes.

Use delivery fakes and BLOB_LOCAL_DIR for development photos. Production must omit fake switches and sign-in-code logging. Database tests and evals erase the database they target.

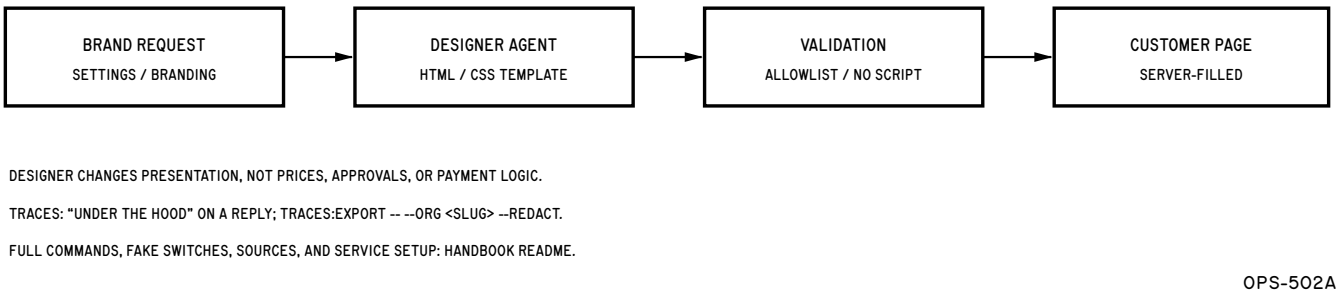


Figure 2.4-2. Customer-Page Designer and Inspection Points

2.4.3 REFERENCE DATA.

Table 2.4-1. Runtime Reference

Parameter	Value
Node.js requirement	24 or newer
Default app origin	http://localhost:3000
Assistant route	/eve/assistant/v1
Designer route, when enabled	/eve/designer/v1
Text queue backstop	Every 1 min
Running-long / follow-ups	Every 10 min / 1 h
Production cron requirement	Vercel Pro

CAUTION

Fake delivery still spends Meta tokens. Offline npm tests force fakes; live evals call the model.

FIRST START	OPERATOR
1 CONFIGURE - key + DB + auth	
2 FAKE - delivery services only	
3 MIGRATE - empty dev database	
4 PROVISION - business + owner	
5 SEED - chosen trade pack	
6 START - npm run dev	
7 SIGN IN - phone / logged code	

OPS